Arista Access Switch Failure SOP: HSR-ACC-27 QoS Congestion Event (ALARM_8649)

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8649

Description

This SOP addresses a Major Severity QoS congestion event on the Arista Access Switch HSR-ACC-27, located at HSR Layout Access, which manifests as Microbursts, interface tail drops, and QoS queue starvation. The underlying cause of this failure domain is attributed to the inherent limitations of the 802.1Q and 802.1p protocols, which are used for QoS classification and policing on this specific role of equipment. The 802.1Q protocol relies on the ingress VLAN tag to determine the QoS priority, whereas the 802.1p protocol relies on the DSCP value within the IP header. However, in a high-traffic scenario, microbursts can occur due to the bursty nature of traffic, leading to interface tail drops and subsequent QoS queue starvation. This, in turn, triggers the ALARM_8649 alarm code, indicating a QoS congestion event.

Pre-requisite Checks

- Pre-requisite: Verify the site location and confirm that the issue is isolated to HSR-ACC-27., Action: Dispatch remote hands to HSR Layout Access to visually inspect the equipment and surrounding infrastructure.

- Pre-requisite: Check the redundant link status between HSR-ACC-27 and the adjacent switch HSR-ACC-26., Action: Verify that the redundant link is operational and not contributing to the congestion event.

Potential Root Causes

Cause	Probability
802.1Q and 802.1p protocol limitations	High
Microbursts and bursty traffic patterns	Medium
QoS queue starvation and interface tail drops	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos queue statistics, Expected Output: Queue statistics for all QoS queues on HSR-ACC-27., Description: This command will provide insight into the current QoS queue utilization and identify potential bottlenecks.

Command: show interface counters, Expected Output: Interface counters for all interfaces on HSR-ACC-27., Description: This command will provide information on the interface traffic patterns and help identify potential sources of congestion.
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```
Command: show qos policy-map, Expected Output: QoS policy map configuration for HSR-ACC-27., Description: This command will provide insight into the QoS policy configuration and help identify potential misconfigurations.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration and correct any misconfigurations., Command: no qos policy-map , Description: This command will delete the incorrect QoS policy map configuration and allow the default configuration to be applied.

Step 2: Step: Adjust QoS queue settings to optimize for high-traffic scenarios., Command: set qos queuing , Description: This command will adjust the QoS queue settings to optimize for high-traffic scenarios, reducing the likelihood of queue starvation and interface tail drops.

Step 3: Step: Verify QoS policy map configuration and validate QoS queue settings., Command: show qos policy-map, Description: This command will verify that the QoS policy map configuration is correct and the QoS queue settings are optimized for high-traffic scenarios.

Post-Check Verification

```
Command: show qos queue statistics, Expected Output: Queue statistics for all QoS queues on HSR-ACC-27, indicating optimized utilization., Description: This command will verify that the QoS queue utilization is optimized and queue starvation is no longer occurring.

Command: show interface counters, Expected Output: Interface counters for all interfaces on HSR-ACC-27, indicating reduced congestion., Description: This command will verify that the interface traffic patterns have been reduced and congestion is no longer occurring.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration., Command: copy running-config startup-config, Description: This command will restore the original QoS policy configuration, ensuring that the system reverts to its previous state.

- Step: Verify QoS policy configuration and correct any misconfigurations., Command: show qos policy-map, Description: This command will verify that the QoS policy configuration is correct and any misconfigurations have been addressed.